

MedLexSeg: Medical Lexicon-Aware Prompt Learning for Abdominal CT Organ Segmentation

Abstract—Text-guided medical image segmentation offers a flexible alternative to fixed-label segmentation by specifying target anatomical structures with natural language prompts. Recent promptable foundation models, such as SAM3, make this paradigm increasingly feasible by supporting textual and spatial prompts. However, directly applying such general models to abdominal CT organ segmentation remains challenging due to low soft-tissue contrast, ambiguous organ boundaries, and large anatomical appearance variations. In this paper, we propose MedLexSeg, a text-guided abdominal CT organ segmentation framework that adapts SAM3 to medical domain through learned text-to-region grounding. MedLexSeg converts short textual organ prompts into reliable box prompts and then uses frozen SAM3 components for mask generation. To improve prompt understanding and localization, MedLexSeg introduces a Medical Lexicon Adapter to refine token-level and prompt-level text representations, a Text-Guided Semantic Enhancement Module to enhance high-level visual features with prompt-specific semantics, and a Multi-Layer Text-Conditioned Deformable Decoder with presence-aware calibration to generate spatial cues for SAM3-based segmentation. Experiments are conducted on FLARE22 and AMOS22 for 13 abdominal organs. MedLexSeg achieves 73.87% mIoU on the in-domain FLARE22 test set and 65.17% mIoU on the out-of-domain AMOS22 test set, outperforming representative text-guided medical segmentation methods, natural-image text-guided segmentation baselines, and a two-stage Grounding DINO+SAM3 baseline. Further evaluations with synonymous and compositional prompts demonstrate improved robustness to textual expression variations. These results suggest that learning medical-domain text-to-region grounding is effective for adapting promptable foundation models to abdominal CT organ segmentation.

Index Terms—Medical image segmentation, text-guided segmentation, SAM3.

I. INTRODUCTION

Abdominal computed tomography (CT) organ segmentation is a fundamental task in medical image analysis, supporting anatomical quantification, radiotherapy planning, surgical navigation, and computer-aided diagnosis. In recent years, deep convolutional networks and vision transformers have achieved remarkable progress in abdominal multi-organ segmentation, with representative methods such as U-Net, nnU-Net, and transformer-based segmentation networks showing strong performance on public benchmarks [1]–[3]. However, most existing methods still follow a fixed-label prediction paradigm, where the model outputs masks for a predefined set of categories. Although effective in closed-set segmentation, this paradigm provides limited flexibility when users need to specify a target organ through textual prompts.

Recent promptable foundation models provide a more flexible segmentation interface. In particular, SAM and its successors have demonstrated strong prompt-driven mask generation ability, and SAM3 further supports concept-level and textual prompts for segmentation [4], [5]. This makes it possible to specify a target structure using text rather than relying solely on fixed output labels. However, SAM3 is not specifically optimized for large-scale medical CT organ segmentation, and its text-to-visual grounding ability may not directly transfer to abdominal CT images. In abdominal CT, anatomical structures are usually represented as low-contrast soft-tissue regions with ambiguous boundaries, large scale variations, and strong appearance similarity among adjacent organs. Large organs such as the liver and spleen occupy broad regions, whereas small or thin structures such as the adrenal glands, esophagus, and duodenum may appear only in limited slices and exhibit weak visual responses. These characteristics make it challenging to precisely anchor a short organ prompt to the corresponding anatomical region in CT images.

Motivated by this observation, we propose **MedLexSeg**, a medical lexicon-aware prompt learning framework for text-guided abdominal CT organ segmentation. MedLexSeg is built upon SAM3, but does not simply rely on its generic textual prompting ability. Instead, it learns to bridge organ-level textual semantics and SAM3-compatible spatial prompting through medical-domain adaptation. Given an abdominal CT image and a textual organ prompt, MedLexSeg first extracts medical lexicon-aware text representations, then enhances high-level CT image features with textual cues, and finally decodes prompt-specific spatial cues to guide SAM3 in generating the target organ mask.

Specifically, MedLexSeg contains three major components. First, a *Medical Lexicon Adapter* refines the frozen text encoder features and produces both token-level and prompt-level text representations, enabling more effective use of short organ prompts. Second, a *Text-Guided Semantic Enhancement Module* injects textual organ cues into high-level image features, encouraging the visual representation to focus on prompt-relevant anatomical regions before spatial prompt decoding. Third, a *Multi-Layer Text-Conditioned Deformable Decoder* progressively generates spatial cues through text-conditioned query initialization, multi-scale visual sampling, and iterative prompt refinement. A presence-aware calibration strategy is further introduced to suppress unreliable responses when the queried organ is absent or weakly visible in a 2D CT slice. In addition, considering that textual prompts may be expressed in different but semantically related forms, MedLexSeg is designed to improve robustness to reasonable

prompt variations. The main contributions of this work are summarized as follows:

- We propose MedLexSeg, a text-guided abdominal CT organ segmentation framework that adapts SAM3 to medical images through learned text-to-region grounding, enabling organ mask generation from short textual prompts.
- We develop a medical lexicon-aware prompt learning strategy, consisting of a Medical Lexicon Adapter, a Text-Guided Semantic Enhancement Module, and a Multi-Layer Text-Conditioned Deformable Decoder, to improve the alignment between textual organ prompts, CT visual features, and spatial box prompts.
- Extensive experiments on FLARE22 and AMOS22 demonstrate that MedLexSeg achieves superior in-domain and out-of-domain segmentation performance over representative text-guided medical segmentation methods, natural-image text-guided segmentation baselines, and a two-stage Grounding DINO+SAM3 baseline, while also showing improved robustness to synonymous and compositional prompts.

II. RELATED WORK

A. Medical Image Segmentation

Medical image segmentation has been extensively studied as a fundamental task for anatomical structure delineation and quantitative image analysis. Early convolutional architectures, represented by U-Net [1], established the encoder-decoder design with skip connections and became a strong baseline for biomedical segmentation. Later, nnU-Net [2] further demonstrated that carefully configured preprocessing, network design, training, and postprocessing pipelines can achieve highly competitive performance across a wide range of medical segmentation tasks. With the development of transformer-based architectures, models such as Swin UNETR [3] introduced hierarchical self-attention and self-supervised pretraining to improve volumetric medical image segmentation. For abdominal CT organ segmentation, large-scale benchmarks such as FLARE22 and AMOS have promoted robust multi-organ segmentation under diverse imaging conditions [6], [7]. Despite their strong performance, most existing segmentation methods follow a fixed-label prediction paradigm, where the output categories are predefined during training. In contrast, MedLexSeg focuses on text-guided CT organ segmentation, where the target organ is specified by a textual prompt rather than solely by a fixed output channel.

B. Vision-Language Models for Medical Imaging

Vision-language models have shown strong potential for learning transferable representations by aligning visual content with textual descriptions. CLIP [8] demonstrated that large-scale image-text contrastive learning can provide powerful zero-shot visual recognition ability in natural images. Inspired by this paradigm, several medical vision-language models have been developed to exploit radiology reports, biomedical captions, or medical knowledge for cross-modal representation learning. For example, GLORIA [9] learned global and

local medical image-text representations from paired radiology reports, MedCLIP [10] explored contrastive learning from unpaired medical images and texts, and BiomedCLIP [11] extended biomedical vision-language pretraining using large-scale image-text pairs collected from biomedical literature. In parallel, promptable segmentation foundation models have introduced a more flexible interface for segmentation. SAM [4] showed strong mask generation ability with visual prompts, while MedSAM [12] adapted this paradigm to medical image segmentation. More recently, SAM3 [5] further supports concept-level prompts, including textual prompts, for segmentation. However, directly transferring generic promptable models to abdominal CT remains challenging because CT organs are often low-contrast, weakly bounded, and highly variable in scale. MedLexSeg therefore aims to enhance SAM3 with medical-domain prompt learning rather than training a new vision-language foundation model from scratch.

C. Text-Driven Medical Image Segmentation

Text-driven medical image segmentation aims to use language information to specify or assist the segmentation target. LViT [13] introduced language information into a vision transformer framework and used text annotations to enhance medical image segmentation, especially under semi-supervised settings. CLIP-driven [14] universal segmentation methods further explored the use of text embeddings to guide organ and tumor segmentation across partially labeled datasets. SAT [15] constructed a large-scale text-prompted medical segmentation model by incorporating anatomical terminology and medical knowledge into the segmentation framework. These methods demonstrate that textual information can provide useful semantic guidance for medical segmentation. However, most existing text-driven methods directly fuse text and visual features to predict segmentation masks. For SAM3-based abdominal CT segmentation, a key issue is how to reliably anchor short organ prompts to the corresponding anatomical regions before mask generation. Different from direct text-to-mask prediction, MedLexSeg explicitly learns prompt-specific spatial cues through medical lexicon adaptation, text-guided semantic enhancement, and deformable prompt decoding, thereby improving the text-to-region grounding process for SAM3-based CT organ segmentation.

III. METHOD

A. Method Overview

The overall framework of **MedLexSeg** is shown in Fig. 1. Given an abdominal CT image $I \in \mathbb{R}^{H \times W \times C}$ and a textual organ prompt T , MedLexSeg aims to segment the anatomical region specified by the prompt. Unlike fixed-label segmentation models that rely on predefined output channels, MedLexSeg performs text-guided segmentation by grounding the input prompt to the corresponding CT region.

MedLexSeg is built upon SAM3 and consists of frozen SAM3 components and a trainable text-guided localization module. The frozen SAM3 image encoder extracts multi-scale image features, and the frozen SAM3 text encoder encodes the input prompt. The trainable part contains three key

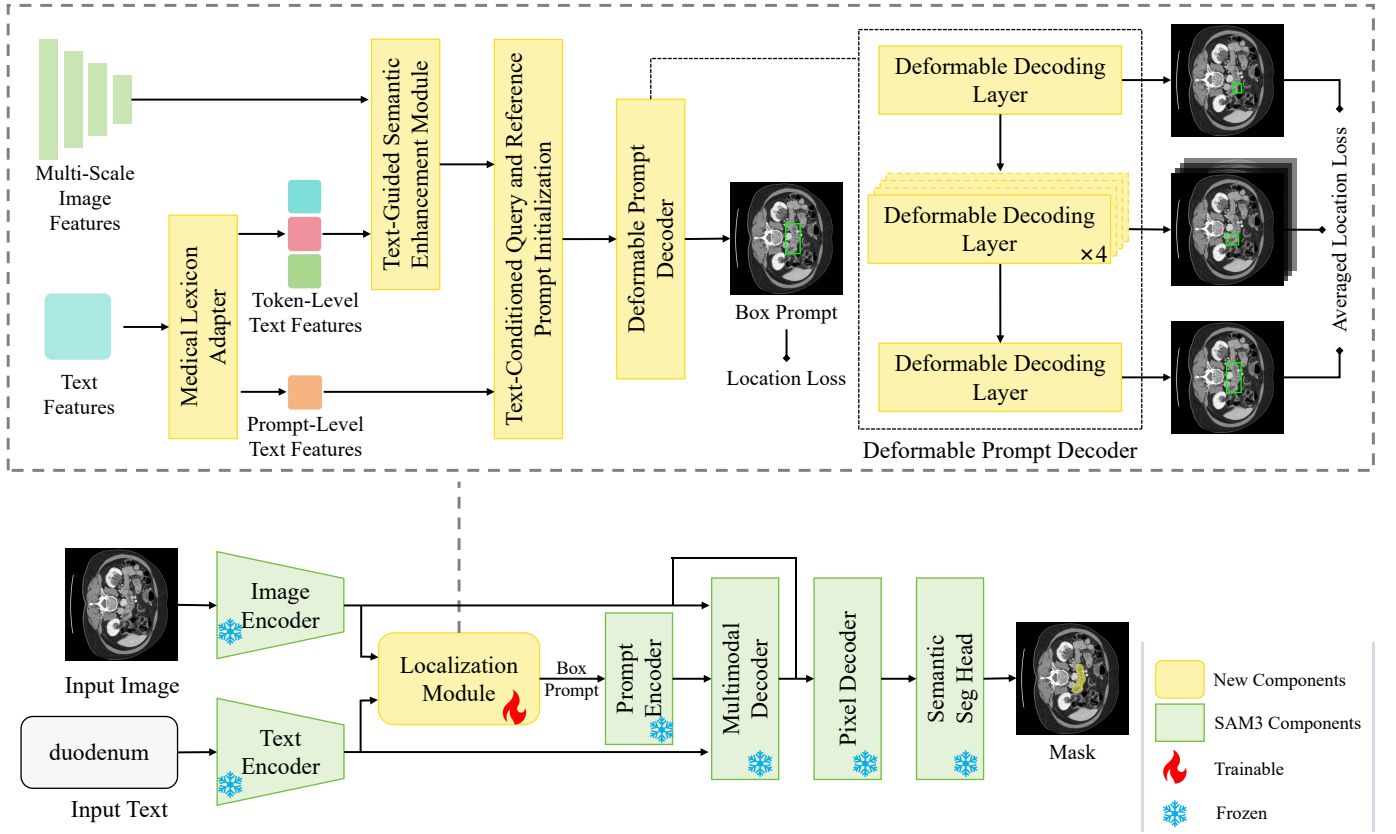


Fig. 1. Overall framework of MedLexSeg. Given an abdominal CT image and a textual organ prompt, frozen SAM3 encoders extract multi-scale image features and text features. The proposed Medical Lexicon Adapter produces token-level and prompt-level text features, while the Text-Guided Semantic Enhancement Module injects prompt-specific semantics into high-level image features. The Multi-Layer Text-Conditioned Deformable Decoder refines text-conditioned queries and reference boxes to generate a box prompt, which is then fed into frozen SAM3 components for mask generation.

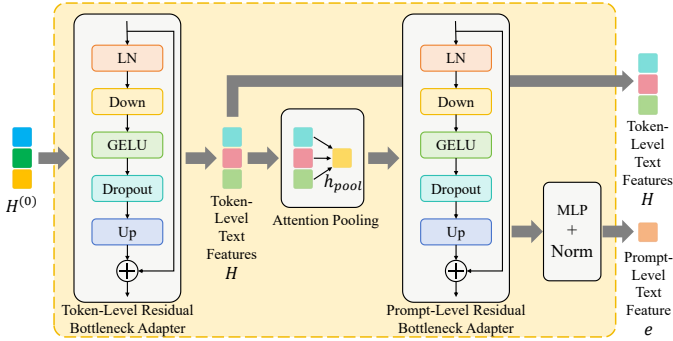


Fig. 2. Illustration of the Medical Lexicon Adapter.

components: a *Medical Lexicon Adapter* for adapting organ-prompt representations, a *Text-Guided Semantic Enhancement Module* for prompt-specific visual feature enhancement, and a *Multi-Layer Text-Conditioned Deformable Decoder* for spatial prompt generation. The predicted box is finally used as the box prompt for SAM3-based mask generation. Thus, MedLexSeg follows a text-to-region-to-mask pipeline, where the localization module provides a medical-domain spatial prior and SAM3 produces the final segmentation mask.

B. Medical Lexicon Adapter

Although the SAM3 text encoder provides general semantic representations, its original text features are not specifically optimized for abdominal CT organ prompts. We therefore introduce a Medical Lexicon Adapter after the frozen text encoder to obtain organ-aware textual representations.

The structure of the Medical Lexicon Adapter is illustrated in Fig. 2. Given a textual prompt T , the frozen SAM3 text encoder outputs the last-layer token features $H^{(0)} = \{h_i^{(0)}\}_{i=1}^L \in \mathbb{R}^{L \times d_t}$, where L is the text sequence length and d_t is the text feature dimension. A token-level residual bottleneck adapter is applied to each token as:

$$h_i = h_i^{(0)} + \gamma_t \text{Up} \left(\text{Dropout} \left(\phi \left(\text{Down} \left(\text{LN} \left(h_i^{(0)} \right) \right) \right) \right) \right), \quad (1)$$

where $\text{LN}(\cdot)$ denotes layer normalization, $\text{Down}(\cdot)$ and $\text{Up}(\cdot)$ are linear projections, $\phi(\cdot)$ denotes GELU activation, and γ_t is a learnable gate. The adapted token-level text features are denoted as $H = \{h_i\}_{i=1}^L$.

To obtain a compact prompt-level feature, we perform attention pooling over H . The attention score of the i -th token is computed as $a_i = w_{\text{pool}}^\top \text{LN}(h_i)$, and the normalized attention weight is defined as:

$$\alpha_i = \frac{\exp(a_i)}{\sum_{j=1}^L \exp(a_j)}, \quad (2)$$

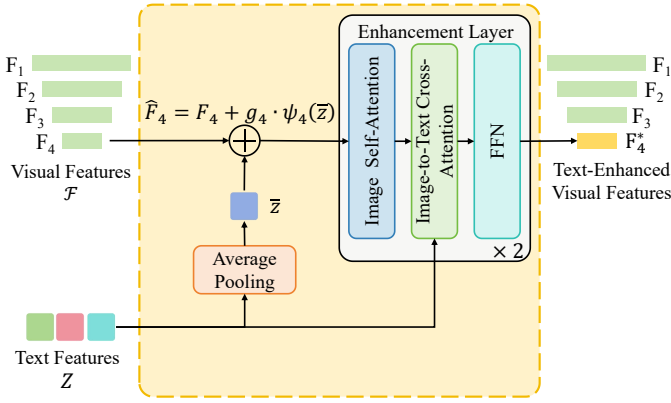


Fig. 3. Illustration of the Text-Guided Semantic Enhancement Module.

where w_{pool} is a learnable pooling vector. The pooled feature is computed as $h_{\text{pool}} = \sum_{i=1}^L \alpha_i h_i$. We further apply a prompt-level residual adapter as:

$$\tilde{h}_{\text{pool}} = h_{\text{pool}} + \gamma_p \text{Up}(\text{Dropout}(\phi(\text{Down}(\text{LN}(h_{\text{pool}}))))), \quad (3)$$

where γ_p is a learnable gate. Finally, a projection head maps the adapted pooled feature into the prompt-level matching space as:

$$e = \text{Norm}(\text{MLP}(\tilde{h}_{\text{pool}})) \in \mathbb{R}^{d_p}, \quad (4)$$

where d_p denotes the prompt-level feature dimension. The adapter outputs token-level text features H for decoder interaction and a prompt-level text feature e for query initialization and matching.

C. Text-Guided Semantic Enhancement Module

The frozen SAM3 image encoder extracts multi-scale visual features, but these features are not explicitly conditioned on the queried organ. To introduce prompt-specific semantics before spatial prompt decoding, we apply a Text-Guided Semantic Enhancement Module to the highest-level visual feature.

The structure of the Text-Guided Semantic Enhancement Module is illustrated in Fig. 3. Let the multi-scale visual features be $\mathcal{F} = \{F_1, F_2, F_3, F_4\}$, where $F_l \in \mathbb{R}^{B \times D \times H_l \times W_l}$ denotes the l -th feature after channel projection and level embedding. The token-level text features are projected into the decoder space as $Z = \{z_i\}_{i=1}^L \in \mathbb{R}^{B \times L \times D}$. We apply semantic enhancement only to F_4 , which provides high-level semantic abstraction while keeping lower-level spatial details unchanged.

We first average-pool valid text tokens to obtain a global text condition vector as:

$$\bar{z} = \frac{1}{L_v} \sum_{i=1}^{L_v} z_i, \quad (5)$$

where L_v is the number of valid tokens. A prompt-conditioned bias is then injected into the highest-level feature as:

$$\hat{F}_4 = F_4 + g_4 \cdot \psi_4(\bar{z}), \quad (6)$$

where $\psi_4(\cdot)$ is a linear projection and g_4 is a learnable scalar gate. The enhanced feature $\hat{F}_4$ is flattened into image tokens

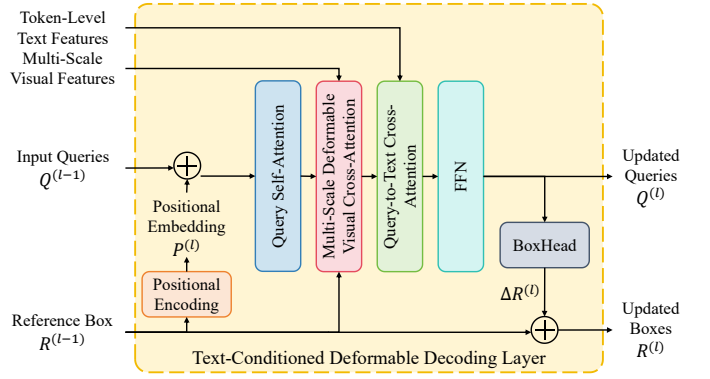


Fig. 4. Illustration of the Text-Conditioned Deformable Decoding Layer.

$X_4 \in \mathbb{R}^{B \times (H_4 W_4) \times D}$. For the m -th enhancement layer, image tokens are updated as:

$$\begin{aligned} \tilde{X}_4^{(m)} &= X_4^{(m-1)} + \text{SelfAttn}(\text{LN}(X_4^{(m-1)})), \\ \bar{X}_4^{(m)} &= \tilde{X}_4^{(m)} + \text{CrossAttn}(\text{LN}(\tilde{X}_4^{(m)}), Z), \\ X_4^{(m)} &= \bar{X}_4^{(m)} + \text{FFN}(\text{LN}(\bar{X}_4^{(m)})), \end{aligned} \quad (7)$$

where $\text{SelfAttn}(\cdot)$, $\text{CrossAttn}(\cdot)$, and $\text{FFN}(\cdot)$ denote image self-attention, image-to-text cross-attention, and feed-forward network, respectively. The updated tokens are reshaped back to the spatial feature map, yielding a multi-scale feature set whose highest-level feature is text-enhanced.

D. Text-Conditioned Query and Reference Box Initialization

Before deformable decoding, MedLexSeg initializes query content with prompt-level textual semantics and assigns each query a spatial reference box. The prompt-level feature e is projected into the decoder space as:

$$c = \text{LN}(W_e e + b_e) \in \mathbb{R}^D, \quad (8)$$

where W_e and b_e are learnable projection parameters. Given M learnable base queries $Q^{\text{base}} = \{q_j^{\text{base}}\}_{j=1}^M \in \mathbb{R}^{M \times D}$, the initial text-conditioned queries are constructed as:

$$q_j^{(0)} = q_j^{\text{base}} + c, \quad j = 1, \dots, M, \quad (9)$$

where $q_j^{(0)}$ denotes the j -th text-conditioned query. The initial query set is denoted as $Q^{(0)} = \{q_j^{(0)}\}_{j=1}^M$.

Each query is associated with an initial reference box $R^{(0)} = \{r_j^{(0)}\}_{j=1}^M$, where $r_j^{(0)} = (c_{x,j}^{(0)}, c_{y,j}^{(0)}, w_j^{(0)}, h_j^{(0)}) \in [0, 1]^4$. The initial reference boxes are distributed over the image plane using a grid-like geometric prior, providing broad spatial coverage for subsequent prompt decoding. During decoding, the reference boxes are iteratively refined in the inverse-sigmoid space to stabilize box updates within the normalized coordinate range.

E. Multi-Layer Text-Conditioned Deformable Decoder

The Multi-Layer Text-Conditioned Deformable Decoder refines query features and reference boxes through iterative

interaction with multi-scale visual features and token-level text features. In our implementation, the decoder consists of six text-conditioned deformable decoding layers with the same structure. The structure of the text-conditioned deformable decoding layers is illustrated in Fig. 4. Let $L_d = 6$ denote the number of decoding layers. For the ℓ -th layer, the input query set and reference box set are $Q^{(\ell-1)}$ and $R^{(\ell-1)}$, respectively. Each reference box $r_j^{(\ell-1)}$ is encoded into a positional embedding $p_j^{(\ell)} = E_{\text{box}}(r_j^{(\ell-1)})$, forming $P^{(\ell)} = \{p_j^{(\ell)}\}_{j=1}^M$.

Each text-conditioned deformable decoding layer contains query self-attention, deformable visual cross-attention, text cross-attention, and feed-forward update, formulated as:

$$\begin{aligned}\tilde{Q}^{(\ell)} &= Q^{(\ell-1)} + \text{SelfAttn}\left(\text{LN}\left(Q^{(\ell-1)} + P^{(\ell)}\right)\right), \\ \bar{Q}^{(\ell)} &= \tilde{Q}^{(\ell)} + \text{DefAttn}\left(\text{LN}\left(\tilde{Q}^{(\ell)} + P^{(\ell)}\right), R^{(\ell-1)}, \mathcal{F}\right), \\ \hat{Q}^{(\ell)} &= \bar{Q}^{(\ell)} + \text{CrossAttn}\left(\text{LN}\left(\bar{Q}^{(\ell)} + P^{(\ell)}\right), Z\right), \\ Q^{(\ell)} &= \hat{Q}^{(\ell)} + \text{FFN}\left(\text{LN}(\hat{Q}^{(\ell)})\right),\end{aligned}\quad (10)$$

where $\text{DefAttn}(\cdot)$ denotes the multi-scale deformable visual cross-attention. Unlike standard global cross-attention that densely attends to all spatial locations, deformable attention adaptively samples a small set of key points from each feature level according to the current query and reference box [16]. In our implementation, it uses 8 attention heads over 4 feature levels, with 4 sampling points per head at each level. The sampling offsets are predicted from query features, and the sampling points are locally generated around the current reference box, enabling efficient aggregation of prompt-relevant visual evidence. $\text{CrossAttn}(\cdot)$ further injects token-level text semantics into query updating.

The reference boxes are refined at each layer. For the j -th query, the box head predicts the coordinate offset as $\Delta r_j^{(\ell)} = \text{BoxHead}^{(\ell)}(q_j^{(\ell)})$. The reference box is updated in the inverse-sigmoid space as:

$$r_j^{(\ell)} = \sigma\left(\Delta r_j^{(\ell)} + \sigma^{-1}\left(r_j^{(\ell-1)}\right)\right), \quad (11)$$

where $\sigma(\cdot)$ is the sigmoid function and $\sigma^{-1}(\cdot)$ denotes its inverse. This iterative update gradually moves candidate boxes toward the target anatomical region.

After the final decoding layer, the query feature $q_j^{(L_d)}$ and the global text condition vector $\tilde{z}$ are projected into a shared matching space as $\tilde{q}_j = W_q q_j^{(L_d)}$ and $\tilde{z} = W_z \tilde{z}$. The text-conditioned classification score is computed as:

$$s_j^{\text{cls}} = \frac{\tilde{q}_j^\top \tilde{z}}{\sqrt{D}}, \quad (12)$$

where $\tilde{q}_j$ and $\tilde{z}$ are the projected query and text features, respectively. To handle slices where the prompted organ is absent or weakly visible, we introduce a presence prediction branch. A learnable presence token is updated in the decoder, and the final presence logit is computed as $s^{\text{pre}} = \text{PresenceHead}(o^{(L_d)})$, where $o^{(L_d)}$ denotes the final presence token. The final ranking score is defined as:

$$s_j = \sigma(s_j^{\text{cls}}) \cdot \sigma(s^{\text{pre}}), \quad (13)$$

where s^{pre} is the presence logit. The predicted box with the highest ranking score is selected as the final box prompt, i.e., $j^* = \arg \max_j s_j$ and $\hat{b} = r_{j^*}^{(L_d)}$.

F. SAM3-Based Box Prompt Segmentation

Given the selected box prompt $\hat{b}$, MedLexSeg uses frozen SAM3 components for box-prompted mask generation as:

$$\hat{M} = \mathcal{S}(I, \hat{b}), \quad (14)$$

where $\mathcal{S}(\cdot)$ denotes the SAM3-based box-prompted segmentation process, and $\hat{M} \in [0, 1]^{H \times W}$ is the predicted mask.

G. Loss Function

Since all SAM3 components are kept frozen, the training objective is imposed only on the text-guided localization module. For each image-text pair (I, T) , we convert the ground-truth mask into a bounding box and use Hungarian matching to assign predicted boxes to target boxes. The localization module is optimized with box regression, generalized IoU, text-conditioned classification, and presence prediction losses.

For the ℓ -th decoding layer, the box regression loss $\mathcal{L}_{\text{bbox}}^{(\ell)}$ supervises the matched box with an L1 distance, and the generalized IoU loss $\mathcal{L}_{\text{giou}}^{(\ell)}$ encourages geometric consistency. Since no fixed-category classification head is used, the text-conditioned classification loss is defined as:

$$\mathcal{L}_{\text{cls}}^{(\ell)} = \frac{1}{M} \sum_{j=1}^M \text{BCE}\left(\sigma(s_j^{\text{cls}}), y_j\right), \quad (15)$$

where $y_j \in \{0, 1\}$ denotes whether the j -th query is matched. The presence loss is defined as:

$$\mathcal{L}_{\text{pre}}^{(\ell)} = \text{BCE}\left(\sigma(s^{\text{pre}}), y^{\text{pre}}\right), \quad (16)$$

where $y^{\text{pre}} \in \{0, 1\}$ indicates whether the prompted organ exists in the current slice.

The detection loss of the ℓ -th layer is formulated as:

$$\mathcal{L}_{\text{det}}^{(\ell)} = \lambda_{\text{bbox}} \mathcal{L}_{\text{bbox}}^{(\ell)} + \lambda_{\text{giou}} \mathcal{L}_{\text{giou}}^{(\ell)} + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}^{(\ell)} + \lambda_{\text{pre}} \mathcal{L}_{\text{pre}}^{(\ell)}, \quad (17)$$

where λ_{bbox} , λ_{giou} , λ_{cls} , and λ_{pre} are loss weights. The total objective is written as:

$$\mathcal{L}_{\text{total}} = \sum_{\ell=1}^{L_d} \mathcal{L}_{\text{det}}^{(\ell)}, \quad (18)$$

where L_d is the number of deformable decoding layers. This layer-wise supervision stabilizes iterative box refinement and improves the quality of the generated box prompt.

IV. EXPERIMENTS

A. Datasets

We evaluate the proposed MedLexSeg on two public abdominal CT datasets, i.e., FLARE22 [6] and AMOS22 [7]. FLARE22 is used for model training and in-domain testing, while AMOS22 is used for out-of-domain testing. Specifically, the FLARE22 training set contains 2492 2D CT slices, and the in-domain test set contains 679 slices. The AMOS22 out-of-domain test set contains 322 slices. All slices are resized to

TABLE I
STATISTICS OF THE ABDOMINAL CT SLICES USED IN OUR EXPERIMENTS.

Organ	Train	In-Domain	Out-Domain
Liver	2335	460	206
Right kidney	1298	438	174
Spleen	1559	253	200
Pancreas	1373	391	80
Aorta	2487	649	50
Inferior vena cava	2471	657	275
Right adrenal gland	746	111	220
Left adrenal gland	833	143	322
Gallbladder	731	130	322
Esophagus	612	74	148
Stomach	1754	287	74
Duodenum	1143	410	84
Left kidney	1357	429	179

512×512 pixels. We select 13 abdominal organs shared by the two datasets as segmentation targets. The number of slices for each organ in the training, in-domain test, and out-of-domain test sets is summarized in Table I.

B. Implementation Details and Evaluation Metrics

1) *Implementation Details*: All experiments are implemented with PyTorch and conducted on four NVIDIA GeForce RTX 3090 GPUs. During training, the batch size is set to 8 per GPU. We optimize the trainable modules of MedLexSeg using AdamW for 50 epochs, with an initial learning rate of 1×10^{-4} and a weight decay of 1×10^{-4} . The learning rate is scheduled by CosineAnnealingLR.

To evaluate the robustness of text-guided segmentation to synonymous prompt expressions, we construct a controlled synonymous-prompt setting. For each organ, a set of training prompts is used during training, and a held-out synonymous prompt is used for testing. During training, one prompt is randomly sampled from the corresponding training prompt set as the textual input. For fair comparison, all competing methods are trained on the same FLARE22 training set using the same random prompt replacement strategy. The prompt setting for the 13 organs is shown in Table II.

2) *Evaluation Metrics*: We use mean Intersection over Union (mIoU) as the primary metric for mask segmentation evaluation. For a predicted binary mask $\hat{M}$ and the corresponding ground-truth mask M , IoU is computed as:

$$\text{IoU} = \frac{|\hat{M} \cap M|}{|\hat{M} \cup M|}, \quad (19)$$

The final mIoU is obtained by averaging IoU over all evaluated organ prompts.

To further evaluate the quality of the text-guided localization stage, we report Top-1 mean box IoU (Top-1 mBoxIoU). Since MedLexSeg uses the highest-ranked predicted box as the final box prompt for SAM3, Top-1 mBoxIoU measures the IoU between the selected box $\hat{b}$ and the ground-truth box b derived from the corresponding mask:

$$\text{Top-1 mBoxIoU} = \frac{1}{N} \sum_{i=1}^N \text{IoU}(\hat{b}_i, b_i), \quad (20)$$

where N denotes the number of evaluated image-prompt pairs.

V. RESULTS AND DISCUSSION

A. Comparisons with the State-of-the-Art Methods

We compare MedLexSeg with representative text-guided medical image segmentation methods, including TGSAM-2 [17], LViT [13], RecLMIS [18], ARSeg [19], and FMISeg [20]. We also include several text-guided segmentation methods from natural images, including SAN [21], MaskCLIP [22], and OVSeg [23]. In addition, we construct a two-stage Grounding DINO+SAM3 baseline, where Grounding DINO [24] first generates box prompts according to the input text, and the predicted boxes are then fed into SAM3 [5] for zero-shot mask generation. This baseline follows a similar text-to-box-to-mask pipeline to MedLexSeg. All competing methods are re-trained on the same FLARE22 training set with the same prompt sampling strategy.

1) *In-Domain Results on FLARE22*: Table III reports the in-domain comparison on the FLARE22 test set under canonical organ prompts. MedLexSeg achieves the best average mIoU of 73.87%, outperforming all competing methods. Compared with the two-stage Grounding DINO+SAM3 baseline, which follows a similar text-to-box-to-mask pipeline, MedLexSeg improves the average mIoU from 70.74% to 73.87%. This indicates that the performance gain is not merely brought by using SAM3 for box-prompted mask generation, but mainly comes from the proposed medical-domain text-guided localization module.

From the perspective of supervision, MedLexSeg and the Grounding DINO+SAM3 baseline optimize the localization stage with box-level supervision. In contrast, the other re-trained segmentation baselines directly learn mask prediction using mask-level ground truth. Compared with mask-level supervision, box-level supervision provides coarser spatial information and does not explicitly describe organ boundaries. Nevertheless, MedLexSeg achieves the best overall performance, suggesting that the proposed text-guided localization module and SAM3-based promptable segmentation framework can achieve strong segmentation accuracy with weaker localization supervision. Moreover, compared with Grounding DINO+SAM3 under a similar text-to-box-to-mask pipeline, MedLexSeg shows a clear advantage, indicating that the learned medical-domain localization module is more suitable for abdominal CT images.

Fig. 5 shows qualitative comparisons on the FLARE22 in-domain test set under canonical organ prompts. MedLexSeg produces more complete and anatomically consistent masks than competing methods. Several baselines tend to miss small structures, confuse adjacent organs, or generate noisy responses outside the target anatomical regions. In contrast, MedLexSeg better preserves the spatial layout of abdominal organs, which is consistent with the quantitative improvements in Table III.

2) *Out-of-Domain Results on AMOS22*: Table IV reports the out-of-domain comparison on the AMOS22 test set under canonical organ prompts. MedLexSeg achieves the highest average mIoU of 65.17%, outperforming all competing methods.

TABLE II
PROMPT SETS USED FOR TRAINING AND HELD-OUT SYNONYMOUS PROMPT EVALUATION.

Organ	Training Prompts	Held-out Synonymous Prompt
Liver	liver; hepatic organ; hepatic tissue	hepatic structure
Right kidney	right kidney; right renal organ; right renal structure	right renal unit
Left kidney	left kidney; left renal organ; left renal structure	left renal unit
Spleen	spleen; splenic organ; splenic tissue	splenic structure
Pancreas	pancreas; pancreatic organ; pancreatic tissue	pancreatic structure
Aorta	aorta; abdominal aorta; aortic vessel	aortic structure
Inferior vena cava	inferior vena cava; inferior caval vein; vena cava	caval vein
Right adrenal gland	right adrenal gland; right adrenal organ; right suprarenal gland	right suprarenal organ
Left adrenal gland	left adrenal gland; left adrenal organ; left suprarenal gland	left suprarenal organ
Gallbladder	gallbladder; gall bladder; biliary organ	biliary structure
Esophagus	esophagus; oesophagus; esophageal structure	esophageal tube
Stomach	stomach; gastric organ; gastric structure	gastric cavity
Duodenum	duodenum; duodenal structure; duodenal segment	proximal duodenum

TABLE III

IN-DOMAIN COMPARISON ON FLARE22 UNDER CANONICAL PROMPTS. RESULTS ARE REPORTED IN mIoU (%). BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. RK: RIGHT KIDNEY; IVC: INFERIOR VENA CAVA; RAG/LAG: RIGHT/LEFT ADRENAL GLAND; LK: LEFT KIDNEY.

Method	Liver	RK	Spleen	Pancreas	Aorta	IVC	RAG	LAG	Gallbladder	Esophagus	Stomach	Duodenum	LK	Avg.
TGSAM-2 (MICCAI'25) [17]	87.92	64.06	78.69	41.47	59.52	45.53	9.62	10.49	62.99	32.71	57.10	29.19	52.29	48.58
LViT (TMI'24) [13]	85.41	68.71	77.57	44.10	74.02	56.30	<u>63.03</u>	38.97	<u>77.55</u>	<u>74.08</u>	51.00	<u>40.09</u>	60.80	62.43
RecLMIS (TMI'25) [18]	68.18	17.22	29.32	13.55	57.35	26.17	5.24	3.08	12.91	24.64	15.29	8.49	23.86	23.48
ARSeg (MICCAI'25) [19]	84.16	63.40	73.24	48.24	72.42	44.37	26.73	29.49	63.08	41.62	42.70	27.21	57.75	51.88
FMISeg (MICCAI'25) [20]	52.65	39.90	11.60	18.86	24.68	19.89	9.72	0.66	6.27	37.50	39.02	23.18	9.79	22.59
SAN (CVPR'23) [21]	91.34	66.63	85.49	41.81	81.81	43.56	3.84	0.00	29.03	34.01	48.61	31.44	64.69	47.87
MaskCLIP (ICML'23) [22]	88.71	16.53	19.09	2.58	0.32	0.00	0.33	0.41	0.50	0.00	0.12	1.44	0.00	10.00
OVSeg (CVPR'23) [23]	70.46	61.57	87.56	33.04	83.19	42.38	43.86	11.32	75.48	47.42	1.49	28.68	56.59	49.46
Grounding DINO+SAM3	89.60	87.04	<u>86.46</u>	<u>57.02</u>	<u>91.86</u>	82.34	66.58	51.70	72.58	79.25	<u>62.49</u>	19.88	<u>72.81</u>	<u>70.74</u>
MedLexSeg (Ours)	<u>90.40</u>	<u>81.79</u>	85.90	64.09	91.93	<u>80.02</u>	57.58	<u>50.34</u>	87.88	72.74	70.59	41.22	85.81	73.87

TABLE IV

OUT-OF-DOMAIN COMPARISON ON AMOS22 UNDER CANONICAL PROMPTS. RESULTS ARE REPORTED IN mIoU (%). BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. RK: RIGHT KIDNEY; IVC: INFERIOR VENA CAVA; RAG/LAG: RIGHT/LEFT ADRENAL GLAND; LK: LEFT KIDNEY.

Method	Liver	RK	Spleen	Pancreas	Aorta	IVC	RAG	LAG	Gallbladder	Esophagus	Stomach	Duodenum	LK	Avg.
TGSAM-2 (MICCAI'25) [17]	73.27	38.38	70.82	22.90	54.04	29.42	2.73	5.26	37.53	18.26	53.30	16.55	45.42	35.99
LViT (TMI'24) [13]	81.90	67.04	<u>81.18</u>	36.85	68.49	52.39	46.03	<u>50.75</u>	<u>56.30</u>	<u>38.03</u>	58.06	34.37	70.89	57.10
RecLMIS (TMI'25) [18]	67.15	17.54	30.59	10.31	47.99	29.77	0.72	0.64	3.81	7.85	15.38	7.71	25.58	20.39
ARSeg (MICCAI'25) [19]	77.12	70.67	73.79	25.37	65.86	38.66	24.15	29.55	44.72	30.89	47.61	19.55	67.35	47.33
FMISeg (MICCAI'25) [20]	60.54	54.20	19.05	21.36	38.39	16.67	14.49	22.76	13.03	17.93	31.01	19.70	35.36	28.06
SAN (CVPR'23) [21]	90.22	64.79	76.42	37.11	69.92	42.66	10.08	0.00	23.58	12.36	76.08	24.59	68.40	45.86
MaskCLIP (ICML'23) [22]	84.55	11.24	21.28	0.74	0.00	0.42	1.64	0.19	1.07	0.00	1.75	0.00	0.00	9.45
OVSeg (CVPR'23) [23]	82.88	59.87	57.02	31.16	60.58	41.94	30.94	24.37	54.56	23.44	11.70	27.31	61.76	43.66
Grounding DINO+SAM3	82.14	<u>81.11</u>	77.72	<u>42.82</u>	87.62	<u>67.96</u>	<u>42.20</u>	51.93	46.67	31.26	49.53	19.54	<u>76.57</u>	<u>58.24</u>
MedLexSeg (Ours)	<u>84.71</u>	85.96	85.77	50.73	<u>87.04</u>	71.53	36.54	37.00	63.05	62.72	<u>64.42</u>	<u>32.20</u>	85.56	65.17

Compared with the trained Grounding DINO+SAM3 baseline, MedLexSeg improves the average mIoU from 58.24% to 65.17%, yielding a 6.93 percentage-point gain. This result shows that the proposed medical-domain localization module is not limited to the FLARE22 distribution, but also improves the transferability of text-guided CT organ segmentation to an external dataset.

Fig. 6 presents qualitative comparisons on the AMOS22 out-of-domain test set. Under the dataset shift between FLARE22 and AMOS22, several competing methods show unstable responses, such as missing target organs, confusing adjacent abdominal structures, or producing incomplete masks. In

contrast, MedLexSeg generates more anatomically consistent segmentation results and better preserves the spatial correspondence between textual organ prompts and CT regions. These visual results are consistent with the quantitative improvements in Table IV, further suggesting that learned medical-domain text-to-region grounding provides more reliable spatial cues for SAM3-based mask generation under out-of-domain conditions.

3) *Results on FLARE22 with Synonymous Prompts*: Table V reports the results on FLARE22 under held-out synonymous prompts. MedLexSeg achieves the best average mIoU of 71.45%, outperforming the second-best LViT by 14.59

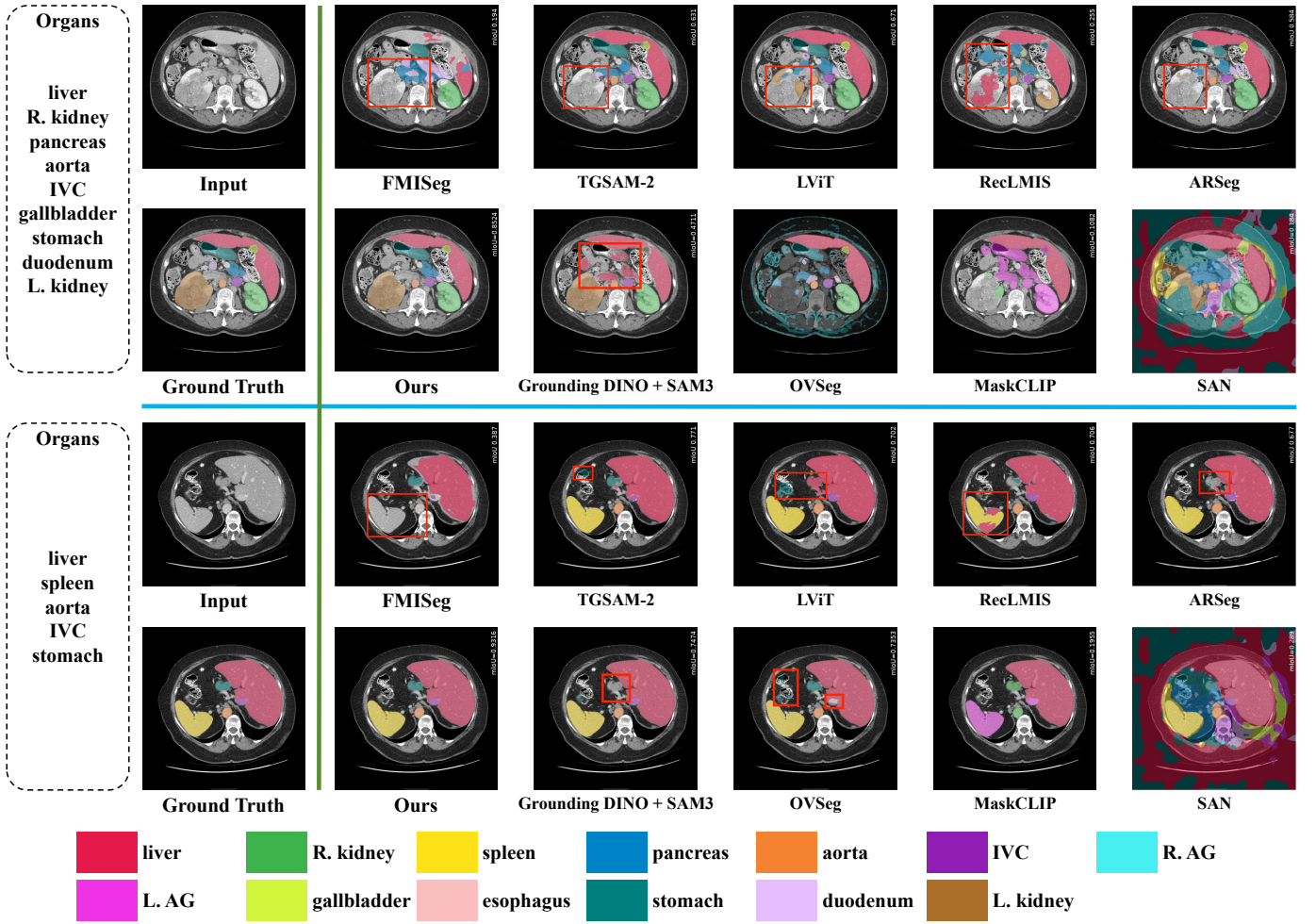


Fig. 5. Qualitative comparison on the FLARE22 in-domain test set under canonical organ prompts.

TABLE V

COMPARISON ON FLARE22 UNDER HELD-OUT SYNONYMOUS PROMPTS. RESULTS ARE REPORTED IN MIOU (%). BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. RK: RIGHT KIDNEY; IVC: INFERIOR VENA CAVA; RAG/LAG: RIGHT/LEFT ADRENAL GLAND; LK: LEFT KIDNEY.

Method	Liver	RK	Spleen	Pancreas	Aorta	IVC	RAG	LAG	Gallbladder	Esophagus	Stomach	Duodenum	LK	Avg.
TGSAM-2 (MICCAI'25) [17]	88.28	64.40	78.32	45.45	57.01	42.86	4.68	5.37	63.70	22.84	<u>56.19</u>	25.68	58.53	47.18
LViT (TMI'24) [13]	85.46	69.85	76.64	44.31	68.30	6.24	65.09	30.23	<u>78.68</u>	<u>72.89</u>	50.68	<u>29.25</u>	61.07	<u>56.86</u>
RecLMIS (TMI'25) [18]	68.37	16.97	23.42	13.23	57.39	26.00	5.82	5.71	12.99	25.13	15.19	6.90	22.95	23.08
ARSeg (MICCAI'25) [19]	84.25	63.57	74.66	<u>48.50</u>	72.34	39.45	28.74	29.48	63.57	41.50	43.11	20.38	57.89	51.34
FMISeg (MICCAI'25) [20]	54.01	42.23	9.81	22.08	28.15	13.18	15.42	0.00	0.35	13.49	32.27	29.38	10.49	20.84
SAN (CVPR'23) [21]	91.83	51.69	17.44	44.36	72.27	11.25	1.63	2.17	17.56	0.00	29.49	28.65	12.64	29.31
MaskCLIP (ICML'23) [22]	59.05	14.38	29.76	9.49	11.40	0.00	0.00	0.00	0.00	1.64	1.74	0.00	0.00	9.80
OVSeg (CVPR'23) [23]	70.03	59.81	87.59	14.63	<u>83.67</u>	41.35	22.08	3.32	6.54	49.87	49.81	28.65	56.97	44.18
Grounding DINO+SAM3	<u>89.92</u>	86.68	72.19	48.32	19.25	69.49	33.06	<u>43.42</u>	4.71	54.01	29.87	2.77	89.70	49.49
MedLexSeg (Ours)	89.54	<u>81.39</u>	<u>85.72</u>	63.82	91.93	<u>65.03</u>	<u>56.76</u>	49.33	87.43	74.38	69.68	27.47	<u>86.31</u>	71.45

percentage points and Grounding DINO+SAM3 by 21.96 percentage points. This result indicates that MedLexSeg is less sensitive to synonymous prompt expressions and can maintain stable segmentation performance when the same target organ is described using alternative but semantically related prompts. Compared with direct text-guided segmentation methods, MedLexSeg benefits from the Medical Lexicon Adapter and text-conditioned localization process, which help map different textual expressions to consistent spatial cues

before SAM3-based mask generation.

The comparison with Grounding DINO+SAM3 is also informative. Although this baseline follows a similar text-to-box-to-mask pipeline and its localization module is trained on the same dataset, its performance decreases notably under synonymous prompts. This suggests that directly adapting a generic grounding model remains sensitive to prompt wording in abdominal CT images. In contrast, MedLexSeg maintains stronger overall performance across synonymous prompts.

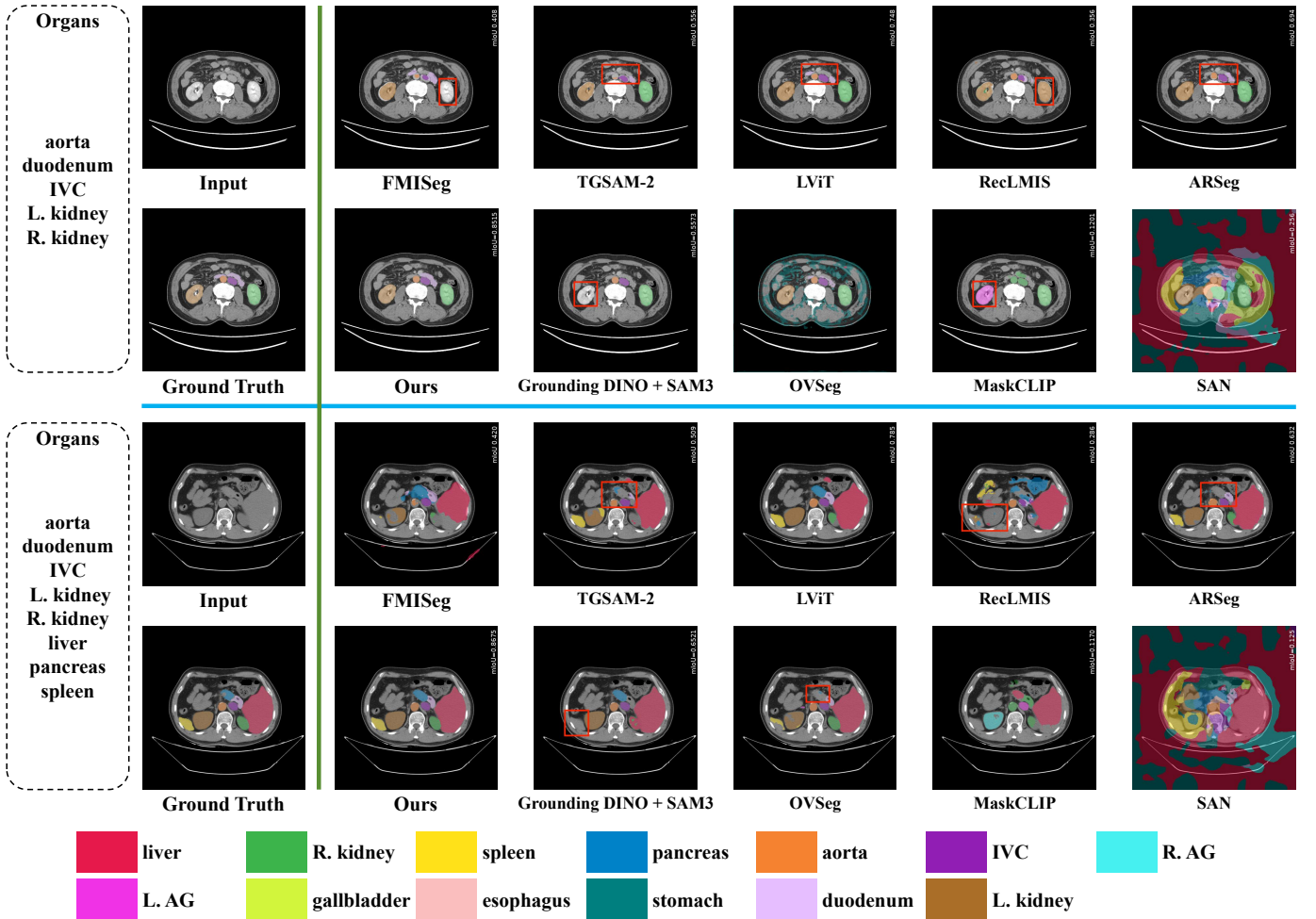


Fig. 6. Qualitative comparison on the AMOS22 out-of-domain test set under canonical organ prompts.

TABLE VI

COMPOSITIONAL PROMPT EVALUATION ON FLARE22. RESULTS ARE REPORTED IN MIOU (%). BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. LK: LEFT KIDNEY; RK: RIGHT KIDNEY; L&R KIDNEY: LEFT AND RIGHT KIDNEY.

Method	LK	RK	L&R Kidney
TGSAM-2 (MICCAI'25) [17]	52.29	64.06	33.99
LVIT (TMI'24) [13]	60.80	68.71	3.44
RecLMIS (TMI'25) [18]	23.86	17.22	26.39
ARSeg (MICCAI'25) [19]	57.75	63.40	6.75
FMISeg (MICCAI'25) [20]	9.79	39.90	0.61
SAN (CVPR'23) [21]	64.69	66.63	39.55
MaskCLIP (ICML'23) [22]	0.00	16.53	5.42
OVSeg (CVPR'23) [23]	56.59	61.57	27.94
Grounding DINO+SAM3	<u>72.81</u>	87.04	<u>56.99</u>
MedLexSeg (Ours)	85.81	<u>81.79</u>	66.90

These results demonstrate that the learned medical-domain localization module improves not only canonical-prompt segmentation but also robustness to synonymous textual expressions, which is important for practical text-guided CT organ segmentation.

4) *Results on FLARE22 with Compositional Prompts*: Table VI reports the results on FLARE22 under the composi-

tional kidney prompt. Different from canonical or synonymous prompts that refer to a single organ, compositional prompts require the model to segment multiple anatomically related structures from one textual expression, such as “left and right kidney”. This setting is more challenging because the model needs to understand the combined semantic target and produce a mask covering both organs. MedLexSeg achieves the best performance under the compositional prompt, reaching 66.90% mIoU for “left and right kidney”, outperforming Grounding DINO+SAM3 and OVSeg by 9.91 and 38.96 percentage points, respectively. This result indicates that MedLexSeg can better preserve the semantic consistency between a compositional text prompt and multiple target regions.

Fig. 7 shows qualitative comparisons under the compositional kidney prompt. Several competing methods either fail to activate both kidneys, generate fragmented masks, or produce false-positive regions outside the target organs. In contrast, MedLexSeg produces more complete bilateral kidney masks and better matches the ground-truth spatial distribution. These results further suggest that the proposed text-to-region grounding strategy is more effective in handling compositional anatomical expressions than directly adapting generic text-guided segmentation or grounding models.

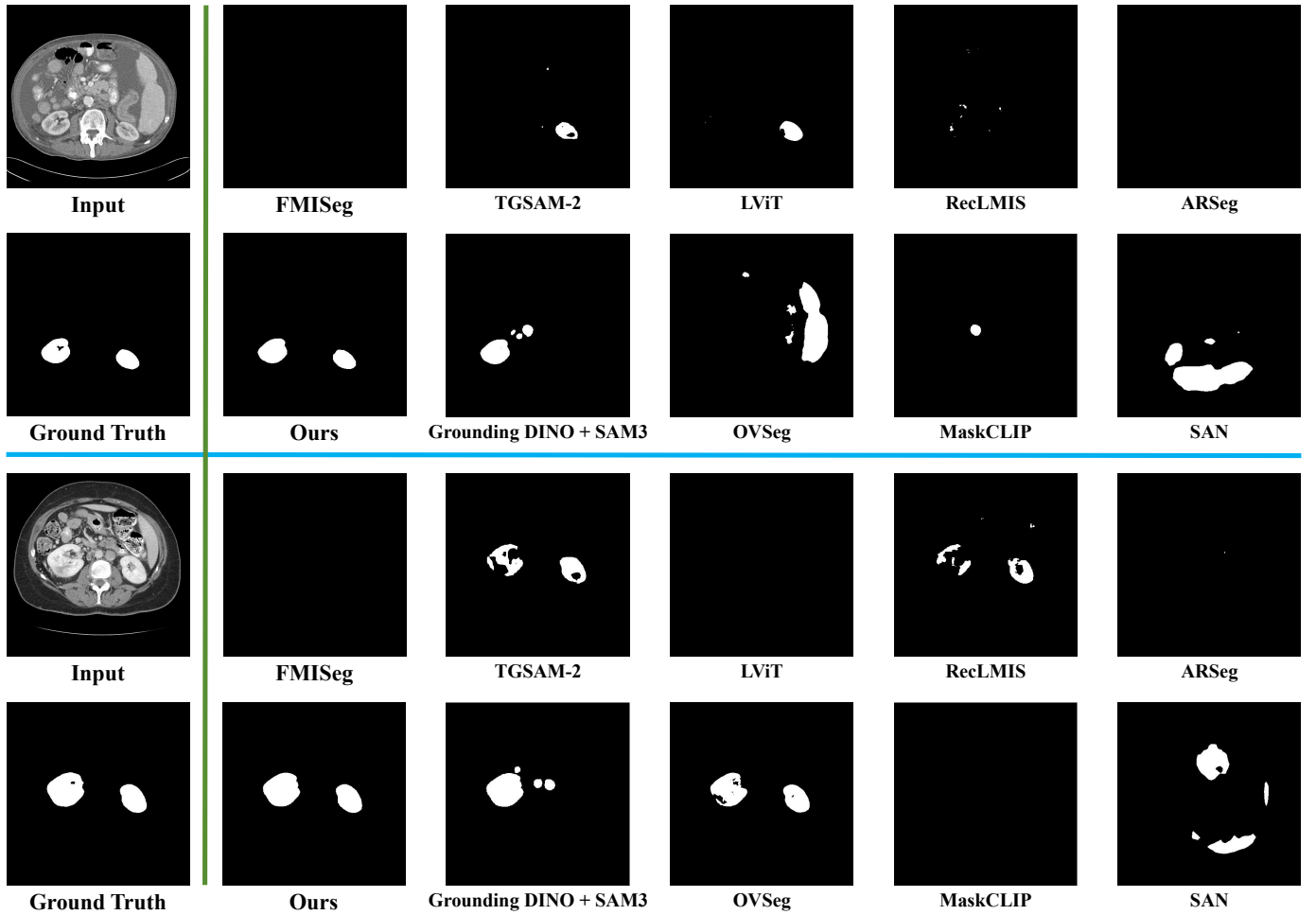


Fig. 7. Qualitative comparison on FLARE22 under the compositional prompt “left and right kidney”. MedLexSeg produces more complete bilateral kidney masks and fewer false-positive regions than competing methods.

TABLE VII

ABLATION STUDY OF THE PROPOSED COMPONENTS ON FLARE22. RESULTS ARE REPORTED IN mIoU (%). MLA: MEDICAL LEXICON ADAPTER; TGSE: TEXT-GUIDED SEMANTIC ENHANCEMENT MODULE.

MLA	TGSE	Canonical	Synonymous
✗	✗	72.53	68.58
✓	✗	72.72	69.28
✗	✓	73.55	69.09
✓	✓	73.87	71.45

B. Ablation Study

1) *Impact of the Proposed Components*: Table VII reports the ablation results of the Medical Lexicon Adapter (MLA) and Text-Guided Semantic Enhancement Module (TGSE) on FLARE22. When both modules are removed, the model obtains 72.53% mIoU under canonical prompts and 68.58% under synonymous prompts. Adding MLA improves the synonymous-prompt result from 68.58% to 69.28%, indicating that adapting text representations helps reduce sensitivity to prompt wording. Adding TGSE improves the canonical-prompt result from 72.53% to 73.55%, showing that text-guided enhancement of high-level image features benefits

spatial prompt generation.

The full MedLexSeg achieves the best performance under both settings, with 73.87% and 71.45% mIoU under canonical and synonymous prompts, respectively. Compared with the variant without MLA, the full model improves the synonymous-prompt result by 2.36 percentage points, suggesting that MLA is important for prompt robustness. Compared with the variant without TGSE, the full model also improves canonical-prompt segmentation, demonstrating that text representation adaptation and prompt-specific visual enhancement are complementary for text-to-region grounding.

2) *Box Localization Quality*: Table VIII evaluates the box localization quality before SAM3-based mask generation. We report Top-1 mBoxIoU, which measures the overlap between the highest-ranked predicted box and the ground-truth box derived from the target mask. Compared with the trained Grounding DINO baseline, MedLexSeg improves Top-1 mBoxIoU from 71.60% to 75.45% under canonical prompts and from 50.56% to 73.00% under held-out synonymous prompts. The larger improvement under synonymous prompts indicates that directly adapting a generic grounding model remains sensitive to prompt wording, whereas MedLexSeg provides more stable medical-domain text-to-region localization.

The ablation results further show that both MLA and TGSE

TABLE VIII

BOX LOCALIZATION QUALITY ON FLARE22. RESULTS ARE REPORTED USING TOP-1 mBoxIoU (%) UNDER CANONICAL AND HELD-OUT SYNONYMOUS PROMPTS. BOLD INDICATES THE BEST RESULT. MLA: MEDICAL LEXICON ADAPTER; TGSE: TEXT-GUIDED SEMANTIC ENHANCEMENT MODULE.

Method	MLA	TGSE	Canonical	Synonymous
Grounding DINO	–	–	71.60	50.56
MedLexSeg	×	×	73.09	68.92
MedLexSeg	×	✓	75.14	70.57
MedLexSeg	✓	×	74.28	71.02
MedLexSeg	✓	✓	75.45	73.00

TABLE IX

EFFECTIVENESS OF DIFFERENT SAM3 PROMPTING STRATEGIES ON FLARE22. RESULTS ARE REPORTED IN mIoU (%). SAM3+TEXT USES ONLY TEXTUAL PROMPTS, WHILE SAM3+GTBOX USES ONLY GROUND-TRUTH BOXES AS ORACLE SPATIAL PROMPTS WITHOUT TEXTUAL INPUT. BOLD INDICATES THE BEST NON-ORACLE RESULT.

Method	Canonical	Synonymous
SAM3+Text	64.88	25.15
SAM3+GTBox	88.11	88.11
MedLexSeg	73.87	71.45

contribute to localization quality. Removing both modules reduces Top-1 mBoxIoU to 73.09% under canonical prompts and 68.92% under synonymous prompts. Adding MLA mainly improves robustness to synonymous prompts, while TGSE enhances prompt-conditioned visual localization by injecting text-specific semantics into high-level image features. The full model achieves the best localization performance in both settings, demonstrating that improved box localization is an important source of the final segmentation gains.

3) *Effectiveness of Learned Box Prompting*: Table IX analyzes the effect of different SAM3 prompting strategies on FLARE22. Directly using SAM3 with textual prompts achieves 64.88% mIoU under canonical prompts, but drops sharply to 25.15% under synonymous prompts. This indicates that the original text-prompting ability of SAM3 is not sufficiently robust for abdominal CT organ segmentation, especially when the prompt expression deviates from canonical organ names.

SAM3+GTBox uses only ground-truth boxes as oracle spatial prompts without textual input. It achieves 88.11% mIoU, showing that SAM3 has strong mask generation capability once accurate spatial prompts are provided. MedLexSeg bridges this gap by learning to convert textual organ prompts into box prompts, achieving 73.87% and 71.45% mIoU under canonical and synonymous prompts, respectively. These results suggest that the main bottleneck is not SAM3-based mask generation itself, but the reliable grounding of textual prompts to anatomical regions. Therefore, learning medical-domain text-to-region grounding is necessary for effectively adapting SAM3 to text-guided CT organ segmentation.

4) *Impact of the Number of Semantic Enhancement Layers*:

Fig. 8 investigates the influence of the number of semantic enhancement layers in the Text-Guided Semantic Enhancement Module (TGSE). Without semantic enhancement layers, the

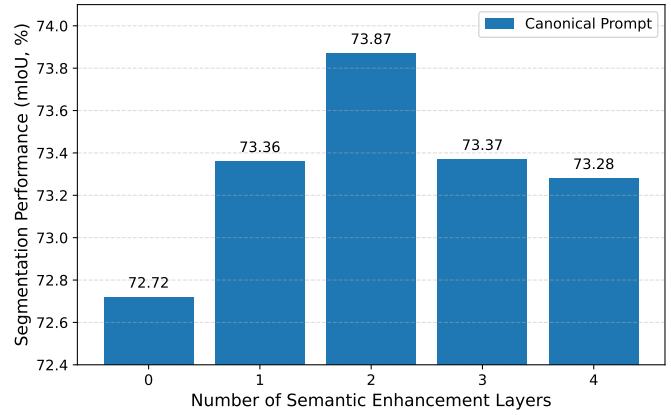


Fig. 8. Ablation study on the number of semantic enhancement layers in TGSE on FLARE22. Results are reported in mIoU (%) under canonical prompts.

TABLE X

ABLATION STUDY OF THE PRESENCE PREDICTION BRANCH ON FLARE22. RESULTS ARE REPORTED IN mIoU (%).

Presence Branch	Canonical	Synonymous
×	73.02	67.90
✓	73.87	71.45

model achieves 72.72% mIoU under canonical prompts. As the number of layers increases from 0 to 2, the performance improves from 72.72% to 73.87%, indicating that moderate image–text interaction can enhance high-level visual features with prompt-specific semantic cues and improve text-to-region grounding for SAM3-based mask generation.

When the number of layers further increases to 3 and 4, the performance decreases to 73.37% and 73.28%, respectively. This suggests that excessive semantic enhancement may introduce redundant cross-modal interactions and disturb the structural information encoded in high-level visual features. Therefore, two semantic enhancement layers provide a better balance between prompt-specific semantic injection and visual representation preservation. Based on this observation, we set the number of semantic enhancement layers to 2 in all other experiments.

5) *Impact of the Presence Prediction Branch*: Table X evaluates the effect of the presence prediction branch on FLARE22. When the presence branch is removed, the model ranks candidate boxes only according to the text-conditioned classification score. This variant obtains 73.02% mIoU under canonical prompts and 67.90% mIoU under synonymous prompts. After introducing the presence branch, the performance increases to 73.87% and 71.45%, yielding improvements of 0.85 and 3.55 percentage points, respectively. These results show that explicit presence modeling brings consistent gains under different prompt settings.

The improvement is more pronounced under synonymous prompts, suggesting that the presence branch helps reduce the influence of prompt-induced semantic shifts. In abdominal CT slices, a prompted organ may be absent, weakly visible, or confused with adjacent structures. Relying only on query–text

matching may therefore select visually plausible but prompt-inconsistent regions. By estimating whether the queried organ exists in the current slice, the presence branch provides an additional prompt-level reliability cue for candidate ranking. This helps suppress prompt-irrelevant responses and improves the stability of text-to-region grounding, leading to more robust SAM3-based mask generation under varied textual expressions.

VI. CONCLUSION

This paper presents MedLexSeg, a text-guided abdominal CT organ segmentation framework that adapts SAM3 to medical image segmentation through learned text-to-region grounding. Instead of directly relying on generic text-prompted segmentation, MedLexSeg converts textual organ prompts into reliable box prompts and then uses frozen SAM3 components for mask generation. Experiments on FLARE22 and AMOS22 demonstrate that MedLexSeg outperforms representative text-guided medical segmentation methods, natural-image text-guided segmentation baselines, and a two-stage Grounding DINO+SAM3 baseline under both in-domain and out-of-domain settings. The results with synonymous and compositional prompts further show that MedLexSeg provides a more stable way to bridge textual organ descriptions and abdominal CT regions, leading to improved robustness under varied prompt expressions. However, the current study is limited to 2D abdominal CT organ segmentation with a relatively fixed set of anatomical categories. Future work will extend MedLexSeg to volumetric 3D segmentation, broader medical modalities, and more clinically diverse textual prompts.

REFERENCES

- [1] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.
- [2] F. Isensee, P. F. Jaeger, S. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnu-net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature methods*, vol. 18, no. 2, pp. 203–211, 2021.
- [3] Y. Tang, D. Yang, W. Li, H. R. Roth, B. Landman, D. Xu, V. Nath, and A. Hatamizadeh, "Self-supervised pre-training of swin transformers for 3d medical image analysis," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 20730–20740.
- [4] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo *et al.*, "Segment anything," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 4015–4026.
- [5] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris, C. Ryali, K. V. Alwala, H. Khedr, A. Huang *et al.*, "Sam 3: Segment anything with concepts," *arXiv preprint arXiv:2511.16719*, 2025.
- [6] J. Ma, Y. Zhang, S. Gu, C. Ge, S. Mae, A. Young, C. Zhu, X. Yang, K. Meng, Z. Huang *et al.*, "Unleashing the strengths of unlabelled data in deep learning-assisted pan-cancer abdominal organ quantification: the flare22 challenge," *The Lancet Digital Health*, vol. 6, no. 11, pp. e815–e826, 2024.
- [7] Y. Ji, H. Bai, C. Ge, J. Yang, Y. Zhu, R. Zhang, Z. Li, L. Zhanng, W. Ma, X. Wan *et al.*, "Amos: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation," *Advances in neural information processing systems*, vol. 35, pp. 36722–36732, 2022.
- [8] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.
- [9] S.-C. Huang, L. Shen, M. P. Lungren, and S. Yeung, "Gloria: A multimodal global-local representation learning framework for label-efficient medical image recognition," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 3942–3951.
- [10] Z. Wang, Z. Wu, D. Agarwal, and J. Sun, "Medclip: Contrastive learning from unpaired medical images and text," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 3876–3887.
- [11] S. Zhang, Y. Xu, N. Usuyama, H. Xu, J. Bagga, R. Tinn, S. Preston, R. Rao, M. Wei, N. Valluri *et al.*, "Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs," *arXiv preprint arXiv:2303.00915*, 2023.
- [12] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, "Segment anything in medical images," *Nature communications*, vol. 15, no. 1, p. 654, 2024.
- [13] Z. Li, Y. Li, Q. Li, P. Wang, D. Guo, L. Lu, D. Jin, Y. Zhang, and Q. Hong, "Lvit: language meets vision transformer in medical image segmentation," *IEEE transactions on medical imaging*, vol. 43, no. 1, pp. 96–107, 2023.
- [14] J. Liu, Y. Zhang, J.-N. Chen, J. Xiao, Y. Lu, B. A. Landman, Y. Yuan, A. Yuille, Y. Tang, and Z. Zhou, "Clip-driven universal model for organ segmentation and tumor detection," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 21152–21164.
- [15] Z. Zhao, Y. Zhang, C. Wu, X. Zhang, Y. Zhang, Y. Wang, and W. Xie, "One model to rule them all: Towards universal segmentation for medical images with text prompts," *arXiv e-prints*, pp. arXiv–2312, 2023.
- [16] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable detr: Deformable transformers for end-to-end object detection," in *International Conference on Learning Representations*, 2021.
- [17] R. Yuan, L. Zhou, J. Xu, Q. Li, M. Chen, Y. Zhang, R. Feng, T. Zhang, and S. Gao, "Tgsam-2: Text-guided medical image segmentation using segment anything model 2," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2025, pp. 565–574.
- [18] X. Huang, H. Li, M. Cao, L. Chen, C. You, and D. An, "Cross-modal conditioned reconstruction for language-guided medical image segmentation," *IEEE Transactions on Medical Imaging*, vol. 44, no. 4, pp. 1821–1835, 2024.
- [19] Q. Wang, X. Lin, and Z. Yan, "Towards robust medical image referring segmentation with incomplete textual prompts," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2025, pp. 636–646.
- [20] B. Yu, J. Yang, Z. Du, Y. Huang, C. Li, and L. Wang, "Frequency-domain multi-modal fusion for language-guided medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2025, pp. 278–288.
- [21] M. Xu, Z. Zhang, F. Wei, H. Hu, and X. Bai, "San: Side adapter network for open-vocabulary semantic segmentation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 12, pp. 15546–15561, 2023.
- [22] Z. Ding, J. Wang, and Z. Tu, "Open-vocabulary universal image segmentation with MaskCLIP," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 8090–8102.
- [23] F. Liang, B. Wu, X. Dai, K. Li, Y. Zhao, H. Zhang, P. Zhang, P. Vajda, and D. Marculescu, "Open-vocabulary semantic segmentation with mask-adapted clip," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 7061–7070.
- [24] S. Liu, Z. Zeng, T. Ren, F. Li, H. Zhang, J. Yang, Q. Jiang, C. Li, J. Yang, H. Su *et al.*, "Grounding dino: Marrying dino with grounded pre-training for open-set object detection," in *European Conference on Computer Vision*. Springer, 2024, pp. 38–55.